

Zhe SHUAI

(+86) 18270613113 | shuaizhe@sjtu.edu.cn

EDUCATION

Shanghai Jiao Tong University (SJTU)

Shanghai, China

Bachelor of Architecture, Minor in Computer Science and Technology

Expected graduation: June 2027

- **GPA:** 90.83/100, **Ranking:** 1/25.
- Relevant Coursework: Data Structures, Operating Systems, Software Engineering, Big Data Algorithms and Analytics, Computer Networks, Machine Learning, etc.
- Awards & Honors: National Scholarship (2024, 2025); Xiaomi Scholarship (2023) ; Gold Award, China International College Students' Innovation Competition (2024); First Prize, National Undergraduate Mathematics Competition (2022) ; First Prize, National Undergraduate Physics Competition (2022).

MITx MicroMasters Program in Statistics and Data Science (SDS)

Remote

Graduate-Level Online Course Program

June 2025 – Present

- Enrolled courses: Probability – The Science of Uncertainty and Data, Machine Learning with Python: from Linear Models to Deep Learning, Fundamentals of Statistics, Data Analysis: Statistical Modeling and Computation in Applications

PUBLICATIONS & PREPRINTS

Conferences

- **Shuai, Z.** *, Xie, X. *, et al. (2026). *BadDreamer: Backdoor attacks against world models for autonomous driving* [Manuscript under review]. *Advances in Neural Information Processing Systems (NeurIPS)*.
- **Shuai, Z.**, Zou, Y., et al. (2027). *FLARE-WAM: Failure-driven latent action refinement for hidden rule adaptation* [Manuscript in preparation for submission to ICLR 2027].
- **Shuai, Z.** *, **Chen, Y.** *, Ni, B., et al. (in preparation). *Video-guided compositional 3D scene reconstruction and in-place completion*[Manuscript in preparation for submission to ICLR 2027].

Journals

- Xie, X., Ran, X., Wu, H., Dong, Z., Xia, B., Jiang, Y., Wang, Z., Sun, J., **Shuai, Z.**, Zhu, A., Zheng, G., Kong, L., & Xue, G. (2026). *CESBench: A multi-task knowledge benchmark for evaluating large language models in construction engineering supervision* [Manuscript submitted for publication]. *IEEE Transactions on Knowledge and Data Engineering*.

Patents

- AI-powered urban carbon emission reduction modeling system (remote-sensing PV identification, emission estimation, ArcGIS integration) , 2024

RESEARCH EXPERIENCES

FLARE-WAM: Failure-Driven World Action Models under Hidden Dynamics Shifts

Shanghai, China

RA | Supervisor: Prof. YingTian Zou, SJTU SAI

Apr. 2026 – Present

- Identified and formulated failure modes in action-conditioned world models caused by hidden dynamics shifts, focusing on robust embodied decision-making.
- Engineered a closed-loop framework integrating latent residuals, inverse action verification, and dynamic candidate reranking to enable real-time belief revision.
- Built and evaluated on VRST-PuzzleGrid, a controlled diagnostic benchmark with OOD-rule and OOD-composition splits, measuring recovery success, candidate reranking accuracy, and shortcut leakage under a strict no-oracle-target protocol.

Vid2Layout: Compositional 3D Scene Reconstruction

Shanghai, China

RA | Supervisor: Prof. BingBing Ni, SJTU Intelligent Media Lab

Apr. 2026 – Present

- Formulated a Video-to-Layout scene understanding task to recover object-centric scene layouts, object relationships, and completable 3D scene representations from casual indoor videos for interactive world model learning.

- Designed and implemented a multi-stage video-to-3D reconstruction pipeline integrating keyframe sampling, open-vocabulary object discovery, cross-frame mask/track association, depth and point-map fusion, and 2D-to-3D grounding to generate stable object sets, local point clouds, and scene graphs.
- Built object-level multi-view partial geometry representations and video-conditioned spatial anchors to mitigate occlusion and sparse-view limitations, improving robustness of object geometry recovery under incomplete observations.
- Explored in-place geometry completion and memory-based training mechanisms inspired by 3D-Fixer and SLAT, with preliminary multi-view geometry ablations showing strong gains over single-view input, including +17.0% mean conditional IoU, +18.7% mean conditional F1, and +6.1% mean bbox IoU on held-out indoor-video scenes.

BadDreamer: Backdoor Attacks on Autonomous Driving Video World Models

Shanghai, China

RA | Supervisor: Prof. GuanjieZheng, SJTU CILAB

Nov. 2025 – May 2026

- Studied data poisoning and backdoor vulnerabilities in autonomous driving video world models, focusing on spatio-temporal trigger injection and semantic erasure attacks for future prediction.
- Designed a highly stealthy trigger-target semantic erasure backdoor pair, using image inpainting to generate supervision for target disappearance in future predicted frames; evaluated low-poisoning-rate fine-tuning and multi-step prediction on driving datasets such as nuScenes.
- Built an end-to-end evaluation pipeline for world-model prediction and planning, achieving 92.5% world-model attack success rate (WM-ASR), 90.3% action attack success rate (Action-ASR), and 86.2% end-to-end attack success with only 5% poisoned data and minimal impact on benign generation quality.

CESBench: Construction Engineering Supervision Benchmark

Shanghai, China

RA | Supervisor: Prof. GuanjieZheng, SJTU CILAB

Oct. 2025 – Feb. 2026

- Designed a specialized data processing pipeline integrating RAG and expert-in-the-loop annotation to extract a high-fidelity dataset from national construction standards and field supervision logs.
- Evaluated the compliance and safety performance of 18 frontier LLMs (including GPT-4o and DeepSeek-V3) in high-stakes construction supervision scenarios, in collaboration with Shanghai Jianke.
- Developed evidence-grounded metrics including Strict Compliance Score (SCS) and Pass@k, revealing up to a 38.1% compliance-performance drop caused by regulation hallucinations.

PROFESSIONAL EXPERIENCES

Apple Inc.,

Shanghai, China

Product Manager Intern

Sept. 2025 – Dec. 2025

- Conducted an AI Glasses industry landscape study, mapping technical roadmaps (Micro-LED, optical solutions) and supply chain dynamics of key players like Ray-Ban Meta and Rokid.
- Led a competitive analysis of AppleCare+ services, comparing Huawei, Xiaomi, Samsung, DJI, and other competitors in pricing ratio, coverage scope, value-added benefits, China/US terms, and repair service fees; delivered a Care Competitive Analysis Report.
- Optimized supply-chain traceability by mapping end-to-end serial number management workflows across hardware production and after-sales service; completed a scan of enterprise application scenarios for AI Agents and spatial computing to support future product opportunity exploration.

Shanghai Jiangnan Technology Co., Ltd. (AI Agent)

Shanghai, China

Co-Founder

Aug. 2024 – Aug. 2025

- Constructed the core interaction paradigm for a vertical-industry AI Agent, managing the full product lifecycle from initial conceptualization to MVP deployment.
- Integrated AI Agent solutions into enterprise marketing and operational workflows, successfully transforming high-frequency business pain points into automated, value-driven capabilities.
- Secured a key commercial deployment with a top-tier pharmaceutical firm, delivering a customized marketing agent that generated CNY 100,000 in early revenue and successfully validated the product-market fit (PMF).
- Spearheaded fundraising efforts resulting in USD 300,000 in angel investment from MiraclePlus, based on the platform's technical scalability and validated commercial potential.

SKILLS

- Programming Languages: Python, C++.
- Technical Stack: PyTorch, TensorFlow, HTML, CSS.